

In this Standard the comma has been adopted as the decimal sign.

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1. SCOPE

This Standard establishes the internal procedures of repairs in composite material parts, typically cosmetics, approved by Embraer's Materials and Structures Engineering's and seeks to formalize some repair practices that may be used directly in the production by means of accompaniment of the Embraer's Manufacturing and Quality Engineering's in order to guarantee the accomplishment of all the requirements of this Standard.

2. APPLICATION

This Standard is applicable to the composite material parts manufactured at Embraer and at subcontracted vendor, since that assist it specified in item 2.1.7.

It is also applied for parts damaged during the handling and/or assembly accomplished in Embraer (origin "Workcenter": Embraer).

NOTES

1) This Standard is valid for parts of the Programs:

- EMB-145 (including Legacy);
- ERJ 170/190 (including Lineage);
- EMB-500/505 (Phenom 100/300); and
- EMB-314 (ALX) fulfilling the applicability specified in items 8 and 9.

2) For Program EMB-550/545 (Legacy 500/450), only repairs procedures described in items 9.6.1, 9.9 and 9.10 are valid.

3) For Program A1M, only repairs procedures described in item 8.13 is valid.

4) For Program EMB-390 (KC-390), only repairs procedures described in items 9.1 to 9.5, 9.6.1, 9.6.2, 9.7 to 9.19 are valid. EMB-390 (KC-390) parts and discrepancy limitation specified in item 9 shall respect and be limited to the same applicability valid for ERJ 170/190.

5) For Program ERJ 190-300/400/500 (E190-E2/E195-E2/E175-E2), only repairs procedures described in items 9.6.1, 9.9 and 9.10 are valid.

2.1 RESTRICTIONS/LIMITATIONS

2.1.1 For repair purpose, the materials may be used out of the white room, since they accomplish the requirements of exhibition control of NE 42-022.

2.1.2 Parts with working temperature above 85°C may not be repaired using this Standard, except in cases described in items 8.9, 9.6.1, 9.9 and 9.10.

2.1.3 Parts and areas that have the Level I classification (NE 20-032) may not be repaired using this Standard, except the cases prescribed in items 8.9, 9.6.1, 9.9 and 9.10.

2.1.4 The hot repairs in the same area shall be limited to two cure cycles.

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NOTES

- 1) In the repair registration document, the cure cycles amount in the repaired area shall be included.
- 2) If the repair was made with hot bonder, new repair in other area with distance according to item 2.1.5 shall be limited according to item 2.1.4.
- 3) If the repair was made in the oven the part shall be limited according to item 2.1.4.

2.1.5 The minimum distance between repairs shall not be smaller than 25,4 mm, counted starting from the repair maximum contour.

2.1.6 For use of this Standard, in Embraer, it is indispensable that the repair is reported in the non-conformity document to guarantee traceability.

2.1.7 For parts to be repaired by the subcontracted vendor it is necessary the emission of non-conformity document that shall obligatorily contains the disposition of the Embraer's Structures, Materials, and/or Interiors Engineering's.

The subcontracted vendor shall suggest the repair type to be accomplished in composite material, according to limitations defined for each repair procedure and may only execute the repair after approval of in the least one of the Engineering (Embraer's Materials, Structures and/or Interiors Engineering).

2.1.8 Repair stations shall be qualified according to NE 20-042.

2.1.9 Approved standard methods of hear application (and controlling) to cure thermosetting resins to perform composite repairs shall be according to NE 20-042.

3. APPLICABLE DOCUMENTS

Embraer Standards

MEP 09-063 - Structural Adhesive Film 121°C (250°F) and 177°C (350°F) - Curing.

MEP 09-066 - Two Components Epoxy Paste and Liquid Adhesive for Bonding.

MEP 10-044 - Finishing Epoxy Mass.

MEP 10-051 - Compound Honeycomb Reinforcement.

MEP 15-006 - Glass Fabrics Preimpregnated with Epoxy Resin 121°C (250°F) - Curing.

MEP 15-008 - Glass Fabrics Preimpregnated with Epoxy Resin 177°C (350°F) - Curing.

MEP 15-016 - Glass Fabrics Preimpregnated with Phenolic Resin, 177°C (350°F) - Curing".

MEP 15-018 - Carbon Fabrics and Tape Preimpregnated with Epoxy Resin, 121°C (250°F) - Curing.

MEP 15-022 - Carbon Fabrics and Tape Preimpregnated with Impact Resistant Epoxy Resin 177°C (350°F) - Curing.

MEP 15-047 - Carbon Fabric and Tape Preimpregnated with Epoxy Resin, 180°C (355°F) - Curing.

MEP 15-067 - Carbon Tape Preimpregnated with Toughened Epoxy Resin, 177°C (350°F) - Curing.

MEP 22-011 - Epoxy Resin Laminating System - Self Extinguishable.

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- NE 20-032 - Composite Materials - Requirements of Parts.
- NE 20-042 - Composite Material - Requirements for Parts Repairs Station
- NE 40-012 - General Parts - Cleaning.
- NE 40-182 - Solid Laminate Curing at 121°C (250°F) - Vacuum Process - Requirements and Manufacturing Procedures.
- NE 40-183 - Solid Laminate Pre-impregnated with Epoxy or Phenolic Resin, Curing at 121°C (250°F) - Autoclave Process - Requirements and Manufacturing Procedures.
- NE 40-189 - Composite Laminate, Curing at 121°C (250°F) / 30PSI - Autoclave Process - Requirements and Manufacturing Procedures.
- NE 40-288 - Carbon Fiber Pre-impregnated with Epoxy Resin, Curing at 132°C (270°F) - Oven or Autoclave Process - Requirements and Manufacturing Procedures.
- NE 42-022 - Prepreg, Dry Fabric with Binder, Adhesive Film, Adhesive Foam, Non-Adhering Film and Adhesive Primer - Transportation, Handling, Storage and Revalidation.
- NE 80-060 - Electrical Bonding - Repair for Composite Materials Parts with Expanded Copper Foil.
- PF 15-001 - Fiberglass/Epoxy Solid Laminates - Manual Impregnation Process.

Foreign Standards

- SAE - AMS - Aerospace Material Specification (SAE - USA)
- SAE - AMS-C-9084 - Cloth, Glass, Finished, for Resin Laminates.
- SAE - AMS 2980 - Carbon Fiber Fabric and Epoxy Resin Wet Lay-Up Repair Material.

4. DEFINITIONS

4.1 DELAMINATION

A separation of plies from each other due to the immigration of resin or mechanical effort.

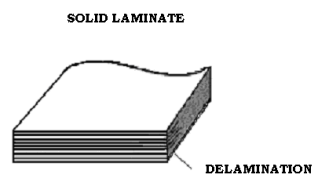


FIGURE 1 : DELAMINATION IN A SOLID LAMINATE

4.2 POROSITY

Occurrence of voids formed by air between the plies or volatile in the laminate resin during the cure process. It may be between plies or just superficially.

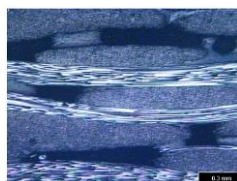


FIGURE 2 : OCCURRENCE OF POROSITY IN A LAMINATE (INCREASING 100X)

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4.3 DISBONDING BETWEEN HONEYCOMB AND SOLID LAMINATE

Breaking of the bonding between the honeycomb and the solid laminate. This bonding is made with resin or adhesive film.

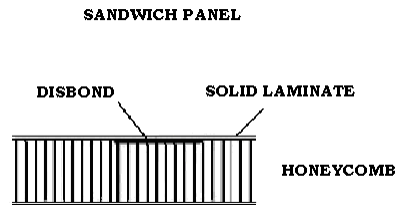


FIGURE 3 : DISBONDING BETWEEN THE HONEYCOMB AND SOLID LAMINATE

4.4 REDUCTION OF THE EDGE DISTANCE

Hole done in place in disagreement with project and excessive machining of edge that take the decreasing of the border distance.

4.5 DISPLACED HOLE

Misalignment of holes in an assembly where is necessary to repair the composite material part. Hole displaced in relation to the project, impeding the assembly.

4.6 IMPROPER HOLE

Executed hole not called on drawing.

4.7 REWORK

Rework procedures are those operations that may be performed by manufacturing personnel to bring previously performed work or work under execution to quality in compliance with Engineering drawing requirements.

4.8 REPAIR

Repair procedures are those operations that are performed to correct the part rejected when a project deviation is acceptable, in the following cases:

- a) damage caused during the transport or subsequent assembly operations; and
- b) correction of part by technical/process different from those used during manufacturing.

5. MATERIALS

The materials to be used in the repairs shall be according to Tables 1 and 2.

TABLE 1 : COLD REPAIR - USED MATERIALS

SPECIFICATION (EMBRAER CODE)	PART NUMBER	MIXING RATIO (PARTS BY WEIGHT)	MATERIAL DESCRIPTION
AMS-C-9084 (E4249305)	PN: GF07781E50100A90000	---	FABRIC, GLASS FIBER - STYLE 7781, CLASS 2, TYPE VIII B
AMS-C-9084 (E4249200)	PN: GF0016D72580A40000	---	FABRIC, GLASS FIBER - STYLE 116, CLASS 2, TYPE II
AMS-C-9084 (E1910357)	PN: 7500-60-CS767	---	FABRIC, GLASS FIBER - STYLE 7500, CLASS 2, TYPE XII A
AMS 2980 (E3120061)	PN: G0904 D 1070 TCT FLAG	---	FABRIC, CARBON FIBER, 3K, PLAIN WEAVE
MEP 09-066 (E1453275)	PN: EA9320NA (2)	100 : 19	EPOXY ADHESIVE, TYPE II
MEP 09-066 (E6578982)	PN: EA9396 (2)	100 : 30	EPOXY ADHESIVE, TYPE III
MEP 10-044 (E7751483)	PN: EPOCAST 7W AT/B (2)	100 : 16	COMPOUND, EPOXY, EDGING FILLER
MEP 10-051 (E1453509) (3)	PN: L-306-12 (2)	100 : 50	EPOXY COMPOUND, MICROSFHERE OF GLASS, TYPE 2, CLASS 3, 0,54 g/cm ³
MEP 10-051 (E1236591)	PN: CG-1305 RESIN/HARDENER (2)	100 : 20	EPOXY RESIN WITH MICROSFHERE OF GLASS, TYPE 2, CLASS 5
MEP 10-051 (E8203537)	PN: EPOCAST 1652-A/B (2)	100 : 12	EPOXY RESIN WITH MICROSFHERE OF GLASS, TYPE 2, CLASS 1
MEP 10-051 (E7798936)	EC-3550 B/A FST 60 LITERS	100 : 52 (4)	EPOXY VOID FILLING COMPOUND, TYPE 2, CLASS 4, 0,57 g/cm ³
MEP 10-51 (E7798986)	EC-3550 B/A FST 380 mL	100 : 52 (4)	EPOXY VOID FILLING COMPOUND, TYPE 2, CLASS 4, 0,57 g/cm ³
MEP 22-011 (E1453498)	PN: EPOCAST 50-A1/9816 (2)	100 : 14	EPOXY RESIN, SELF- EXTINGUISHABLE - LONG WORK TIME (1)
MEP 22-011 (E1453497)	PN: EPOCAST 50-A1/946 (2)	100 : 15	EPOXY RESIN, SELF- EXTINGUISHABLE - SHORT WORK TIME (1)

(1) Work time is the period of resin preparation, lamination of the repair until the finish.

(2) For detailed curing cycles, consult the respective MEP of each resin system.

(3) Deactivated for new drawings from 05 Sep 2024.

(4) 100 parts of component B and 52 parts of component A (in weight).

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TABLE 2 : HOT REPAIR - USED MATERIALS

ORIGINAL PART MATERIAL	ORIGINAL PART MATERIAL DESCRIPTION AND PART NUMBER	REPAIR MATERIAL	ADHESIVE FILM	REPAIR TEMPERATURE (°C)	PROCEDURE
MEP 15-006 (E3659021)	FABRIC, PREPREG, GLASS/EPOXY-STYLE 7781 PN: P1175	E3659021	MEP 09-063 (E2086261)	121	"HOT BONDER"/ VACUUM/OR OVEN
MEP 15-006 (E3670732)	FABRIC, PREPREG, GLASS/EPOXY-STYLE 220 PN: P1176	E3670732	MEP 09-063 (E2086261)	121	"HOT BONDER"/ VACUUM/OR OVEN
MEP 15-008 (E1910513)	FABRIC, PREPREG, GLASS/EPOXY-STYLE 7781 (F161/33%/7781) PN: P1178	E3659021	MEP 09-063 (E2086261)	121	"HOT BONDER"/ VACUUM/OR OVEN
MEP 15-016 (E9125613)	FABRIC, PREPREG, GLASS/PHENOLIC-STYLE 7781 (7781-38-F120-219) PN: B021032	E9125613	NOT APPLICABLE	177	AUTOCLAVE/VACUUM/ 100 psi
MEP 15-018 (E3671097)	FABRIC, PREPREG, CARBON/EPOXY-CLASS 1, STYLE 8HS PN: P1180	E3671097	MEP 09-063 (E2086261)	121	"HOT BONDER"/ VACUUM/OR OVEN
MEP 15-018 (E7725616)	FABRIC, PREPREG, CARBON/EPOXY-CLASS 1, STYLE PLAIN WEAVE PN: P1181E	E7725616	MEP 09-063 (E2086261)	121	"HOT BONDER"/ VACUUM/OR OVEN
MEP 15-022 (E2070720)	FABRIC, PREPREG, CARBON/EPOXY-TYPE 2, CLASS 1, 8HS (F4G-584-F584) PN: P1140	E3671097	MEP 09-063 (E2086261)	121	"HOT BONDER"/ VACUUM/OR OVEN
MEP 15-022 (E2192954)	CARBON/EPOXY-TYPE 1, CLASS 1, 8HS (F4G 584-F584-RC34) PN: P1157	E3671097	MEP 09-063 (E2086261)	121	"HOT BONDER"/ VACUUM/OR OVEN
MEP 15-047 E7724212	FABRIC, PREPREG, CARBON/EPOXY RESIN-CLASS I (8552S/40%/193PW/AS4-3K) PN: P1142E	E7725616	MEP 09-063 (E2086261)	121	"HOT BONDER"/ VACUUM/OR OVEN
MEP 15-067 (E5294499)	TAPE, PREPREG, CARBON/EPOXY RESIN-CLASS 1, PN: P2352W-19L-150	E7725616	MEP 09-063 (E2086261)	121	"HOT BONDER"/ VACUUM/OR OVEN
MH-TCSPF-T-2511 (E8688354)	CARBON FIBER, EPOXY RESIN, FORM 2, TYPE 42, GRADE 190, CLASS 2, STYLE 12K PN: F6273C-11M	E8688354	MEP 09-063 (E2086261)	132	"HOT BONDER"/ VACUUM/OR OVEN

6. VACUUM APPLICATION AND CURING CYCLE

The cure preparation and vacuum application shall be according to NE 40-182.

The cure cycle shall be followed integrally, being any discrepancy reason for repair rejection.

NOTE

For E8688354 material, vacuum bag procedure shall be according to NE 40-182 only for 2 plies, maximum. For 3 plies or righter, apply at least 5 h debulking before curing, according to NE 40-288.

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7. REQUIREMENTS FOR REPAIR

7.1 The repairs established in this Standard shall only be executed by qualified/certified personnel by Embraer Quality according to applicable documents.

7.2 Before the repair execution, the area to be repaired shall be cleaned with free lint, clean cloth soaked with solvent specified by NE 40-012 and before the solvent evaporates, rub dry and clean cloth to remove the residues. If the repair is not executed immediately after the cleaning, the prepared area shall be protected from dust, foreign objects, finger marks.

7.3 All carbon fiber parts shall be checked by ultrasonic inspection to verify its integrity, except the repairs of disbonding/delamination/breaking/displacement or lack of the first or last ply of glass fiber of carbon fiber parts indicated in items 8.9 and 9.9.

8. STANDARD PROCEDURES FOR COSMETIC REPAIRS (NON-STRUCTURAL) IN INTERIOR PARTS AND SEGREGATION WALLS

NOTE

In the electrical bonding areas, ensure the repair condition according to NE 80-060.

8.1 PLY BREAKING (CRACKS)

The crack repair and damage due to the tool entrance is allowed in parts, fulfilling the limitations indicated in Table 3.

TABLE 3 : PLY BREAKING (CRACKS)

PARTS	DISCREPANCY LIMITATIONS
1. CABIN OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 2. NON-STRUCTURAL PARTS OF INTERIORS. 3. SEGREGATION WALLS OF THE PROGRAMS EMB-145, INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 4. FURNISHINGS (DIVIDERS, WARDROBES, TOILETS), CABIN PAX LINERS (SIDE WALL, SIDE LEDGE) AND COCKPIT OF THE PROGRAMS EMB-500/505 (PHENOM 100/300). OBS: NOT APPLICABLE FOR CEILING PANELS, FLOOR PANELS, COCKPIT MAIN PANEL, CONTROL PEDESTAL AND GLARESHIELD OF THE PROGRAMS EMB-500/505 (PHENOM 100/300).	1. MAXIMUM DIMENSION OF 50 x 50 mm. 2. AT MAXIMUM 4 AFFECTED FABRIC PLIES. OBS: NOT APPLICABLE FOR REGIONS OF PART ATTACHMENT. ENSURE MINIMUM DISTANCE OF 25,4 mm (1 in) BETWEEN THE DAMAGE AND THE FASTENERS. 3. MANUFACTURED PARTS WITH MATERIAL E8353578 (HT93 PHENOLIC), LIMIT OF 5% OF THE PART TOTAL AREA.

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REPAIR:

- a) the damage shall be totally eliminated by the sanding (sandpapers grit 60 and/or up to 180) of the affected region, maintaining a staggering of 15 mm (reference);
- b) if all the plies have been affected (for parts with up to 4 plies), one ply shall be left as substratum;
- c) the original lay-up shall be recomposed with fabric (according to Table 1) and Epocast 50-AI resin with overlap of 15 mm;
- d) one ply of the same material, with orientation of $\pm 45^\circ$ and with overlap of 20 mm shall be added on the repair;
- e) the repair shall be cured for 150 min at 65°C according to PF 15-001; and
- f) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

NOTE

If the sanding is not necessary, apply the items "c", "d" and "f" only.

This repair procedure is allowed in the case of the last ply breaking during the procedure of inserts installation "shells" panels.

8.2 MATERIAL LACK (CLEARANCE AND STEPS)

The gap repair, material lack and steps are allowed in parts, fulfilling all the limitations indicated in Table 4.

TABLE 4 : MATERIAL LACK (CLEARANCE AND STEPS)

PARTS	DISCREPANCY LIMITATIONS
1. CABIN OF THE PROGRAMS EMB-145 INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX).	
2. FURNISHINGS (DIVIDERS, WARDROBES, TOILETS), CABIN PAX LINERS (SIDE WALL, SIDE LEDGE) AND COCKPIT OF THE PROGRAMS EMB-500/505 (PHENOM 100/300). OBS: NOT APPLICABLE FOR CEILING PANELS, FLOOR PANELS, COCKPIT MAIN PANEL, CONTROL PEDESTAL AND GLARESHIELD OF THE PROGRAMS EMB-500/505 (PHENOM 100/300).	1. MAXIMUM WIDTH OF 6,0 mm. 2. MAXIMUM LENGTH OF 100 mm. 3. MANUFACTURED PARTS WITH MATERIAL E8353578 (HT93 PHENOLIC), LIMIT OF 5% OF THE PART TOTAL AREA.

The parts shall be inspected in relation to the discrepancy type and the repair procedure shall be selected from 8.2.1 to 8.2.4 in such way to correct the discrepant part.

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8.2.1 Clearance due to Material Lack in the Parts Edge

REPAIR 1:

- a) shall be certified that the edge distance is according to design;
- b) make just chamfers in the two laminate sides in the discrepant regions, according to Figure 4, in order to create anchorage of the resin + chopped glass fiber;
- c) the system line of the part shall be recomposed with resin Epocast 50-A1 and chopped glass fiber;
- d) the repair shall be cured for 150 min at 65°C according to PF 15-001;
- e) the part shall be inspected and adjusted according to assembling necessity, such way that is obtained minimum design clearance;
- f) laminate on the external side passing for the internal side two fabric plies (according to Table 1) and resin Epocast 50-A1, with orientation of $\pm 45^\circ$, overlap of 15 mm for first ply and 20 mm for second ply;
- g) the laminate shall be cured for 150 min at 65°C according to PF 15-001, compacting such way to avoid steps; and
- h) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

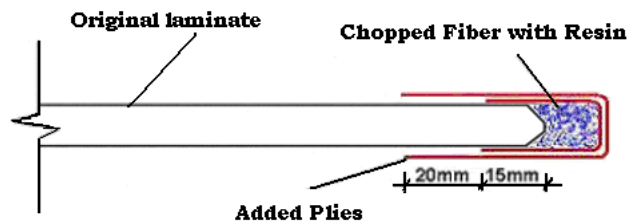


FIGURE 4 : REPAIR OF REGION WITH MATERIAL LACK

REPAIR 2:

Parts with insufficient thickness for application of chopped glass fiber:

- a) shall be verified that the edge distance is according to design and the system line of the part shall be recomposed, laminating plies of glass fiber fabric 7781 (E1910363) and resin Epocast 50-A1 with orientation of $\pm 45^\circ$, and with overlap of 20 mm;
- b) the laminate shall be cured for 150 min at 65°C according to PF 15-001, compacting such way to avoid steps; and
- c) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

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8.2.2 Clearance due to Material Lack in the Region of Part Radius

REPAIR:

- a) the system line of the part shall be recomposed with resin L-306-12 or EC-3550 B/A FST (applicable to all programs) or with resin Epocast 1652 (not applicable to the programs EMB-500/505);
- b) the repair shall be cured for 90 min at 65°C when resin L-306-12 or EC-3550 B/A FST is applied or for 2 h at 65°C when resin Epocast 1652 is applied;
- c) the repair shall be inspected visually according to NE 20-032 requirements, Level III;
- d) the repaired part shall be adjusted according to assembling necessity, so that is obtained the design geometry;
- e) one ply of glass fiber fabric 7781 (E1910363) and resin Epocast 50-A1 shall be laminated on the repair, orientation of $\pm 45^\circ$, according to Figure 5;
- f) the laminated shall be cured for 150 min at 65°C according to PF 15-001; and
- g) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

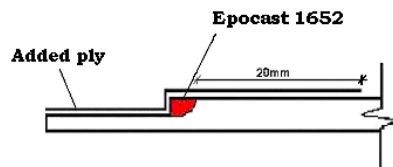


FIGURE 5 : REGION REPAIR WITH MATERIAL LACK

8.2.3 Steps

REPAIR:

- a) so much fabric plies shall be laminated (according to Table 1) and resin Epocast 50-A1 so necessary to obtain the design thickness, alternating orientation of $\pm 45^\circ$ and 0, 90°;
- b) the laminates shall be cured for 150 min at 65°C according to PF 15-001;
- c) after the cure, laminated area shall be sanded (sandpapers grit 60 and/or up to 180) such as to adjust according to necessity;
- d) one ply the same material shall be laminated as finish and the cure process shall be repeated according to item "b" (see Figure 6);
- e) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

NOTE

If applicable, the filling plies may be directly laminated according to item "a" and the reinforcement ply according to item "d" and realize only one cure process according to item "b".

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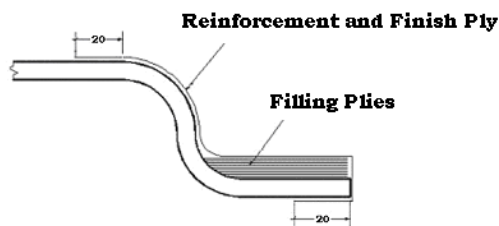


FIGURE 6 : STEPS

8.2.4 Material Lack due to Honeycomb Displacement in the Edges (Parts with up to 2 plies)

The repair of material due to honeycomb displacement in the edges is permitted in parts, fulfilling all the limitations indicated in Table 5.

TABLE 5 : MATERIAL LACK DUE TO HONEYCOMB DISPLACEMENT IN THE EDGES

PARTS	DISCREPANCY LIMITATIONS
1. CABIN OF THE PROGRAMS EMB-145, (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 2. FURNISHINGS (DIVIDERS, WARDROBES, TOILETS), CABIN PAX LINERS (SIDE WALL, SIDE LEDGE) AND COCKPIT OF THE PROGRAMS EMB-500/505 (PHENOM 100/300). OBS: NOT APPLICABLE FOR CEILING PANELS, FLOOR PANELS, COCKPIT MAIN PANEL, CONTROL PEDESTAL AND GLARESHIELD OF THE PROGRAMS EMB-500/505 (PHENOM 100/300).	1. FOR PARTS WITH UP TO 2 PLIES. 2. MANUFACTURED PARTS WITH MATERIAL E8353578 (HT93 PHENOLIC), LIMIT OF 5% OF THE PART TOTAL AREA.

REPAIR:

a) shall be certified that the edge distance is according to design. The system line of the part shall be recomposed with resin L-306-12 or EC-3550 B/A FST (applicable to all programs) or with resin Epocast 1652 (not applicable to the programs EMB-500/505);

b) the repair shall be cured for 90 min at 65°C when resin L-306-12 or EC-3550 B/A FST is applied or for 2 h at 65°C when resin Epocast 1652 is applied;

c) a fabric ply (according to Table 1) and resin Epocast 50-A1 shall be laminated on the external side and on the internal side, with orientation of $\pm 45^\circ$, and with overlap of 20 mm;

d) the laminate shall be cured for 150 min at 65°C according to PF 15-001, compacting in way to avoid steps; and

e) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

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8.3 DELAMINATION

The delamination repair is allowed in parts, fulfilling all the limitations indicated in Table 6.

TABLE 6 : DELAMINATION REPAIR ALLOWED

PARTS	DISCREPANCY LIMITATIONS
1. CABIN OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 2. NON-STRUCTURAL PARTS OF INTERIORS. 3. SEGREGATION WALL OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 4. FURNISHINGS (DIVIDERS, WARDROBES, TOILETS), CABIN PAX LINERS (SIDE WALL, SIDE LEDGE) AND COCKPIT OF THE PROGRAMS EMB-500/505 (PHENOM 100/300). OBS: NOT APPLICABLE FOR CEILING PANELS, FLOOR PANELS, COCKPIT MAIN PANEL, CONTROL PEDESTAL AND GLARESHIELD OF THE PROGRAMS EMB-500/505 (PHENOM 100/300).	1. MAXIMUM DIMENSION OF 50 x 50 mm. 2. AT MAXIMUM 4 PLIES OF AFFECTED FABRIC. OBS: IT IS NOT APPLICABLE FOR REGIONS OF ATTACHMENT OF PARTS. ENSURE MINIMUM DISTANCE OF 25,4 mm (1 in) BETWEEN THE DAMAGE AND THE FASTENERS. 3. MANUFACTURED PARTS WITH MATERIAL E8353578 (HT93 PHENOLIC), LIMIT OF 5% OF THE PART TOTAL AREA.

REPAIR 1:

- a) the delamination in the affected area shall be removed with sanding (sandpapers grit 60 and/or up to 180). A staggering of 15 mm shall be used or to leave a ply as substratum;
- b) the lay-up of design shall be reconstituted with glass fiber fabric 7781 (E1910363) and resin Epocast 50-AI, with design orientation and overlap of 15 mm;
- c) a reinforcement ply of the same fabric shall be added on the area, with orientation of $\pm 45^\circ$ and overlap of 25 mm;
- d) the repair shall be cured for 150 min at 65°C, according to PF 15-001; and
- e) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

REPAIR 2:

- a) inject resin Epocast 50-AI in the delaminated region;
- b) cure for 150 min at 65°C according to PF 15-001; compact the region during the cure; and
- c) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

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8.4 DISPLACED OR OUT-OF ROUND HOLE

The repair of displaced or out of round hole is allowed in parts, fulfilling all the limitations indicated in Table 7.

TABLE 7 : DISPLACED OR OUT-OF ROUND HOLE

PARTS	DISCREPANCY LIMITATIONS
1. CABIN OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 2. NON-STRUCTURAL PARTS OF INTERIORS. 3. SEGREGATION OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 4. FURNISHINGS (DIVIDERS, WARDROBES, TOILETS), CABIN PAX LINERS (SIDE WALL, SIDE LEDGE) AND COCKPIT OF THE PROGRAMS EMB-500/505 (PHENOM 100/300). OBS: NOT APPLICABLE FOR CEILING PANELS, FLOOR PANELS, COCKPIT MAIN PANEL, CONTROL PEDESTAL AND GLARESHIELD OF THE PROGRAMS EMB-500/505 (PHENOM 100/300).	1. MAXIMUM DE 3,0 mm OF DISPLACEMENT. 2. MAXIMUM OF 1 HOLE AT EACH SEQUENCY OF 5 HOLES. 3. MANUFACTURED PARTS WITH MATERIAL E8353578 (HT93 PHENOLIC), LIMIT OF 5% OF THE PART TOTAL AREA.

REPAIR:

- the affected hole shall be filled with resin Epocast 50-AI + chopped glass fiber;
- soon after, a fabric ply shall be laminated (according to Table 1) + resin Epocast 50-AI, with orientation of $\pm 45^\circ$ and overlap of 25 mm, on both sides of the part, on the covered hole;
- the laminate shall be cured for 150 min at 65°C according to PF 15-001; and
- the repair shall be inspected visually according to NE 20-032 requirements, Level III.

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8.5 IMPROPER HOLE OR DRILL MARKS

The repair of improper holes or drill marks is allowed in parts, fulfilling all the limitations indicated in Table 8.

TABLE 8 : IMPROPER HOLE OR DRILL MARKS

PARTS	DISCREPANCY LIMITATIONS
1. CABIN OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 2. NON-STRUCTURAL PARTS OF INTERIORS. 3. SEGREGATION WALL OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 4. FURNISHINGS (DIVIDERS, WARDROBES, TOILETS), CABIN PAX LINERS (SIDE WALL, SIDE LEDGE) AND COCKPIT OF THE PROGRAMS EMB-500/505 (PHENOM 100/300). OBS: NOT APPLICABLE FOR CEILING PANELS, FLOOR PANELS, COCKPIT MAIN PANEL, CONTROL PEDESTAL AND GLARESHIELD OF THE PROGRAMS EMB-500/505 (PHENOM 100/300).	1. MAXIMUM DIAMETER OF HOLE: 6,4 mm (1/4in). 2. MINIMUM DISTANCE OF EDGE: 2,5d AND PITCH OF 4D mm. d = diameter of improper hole D= diameter of greater hole 3. MANUFACTURED PARTS WITH MATERIAL E8353578 (HT93 PHENOLIC), LIMIT OF 5% OF THE PART TOTAL AREA.

REPAIR:

- the affected hole shall be filled with resin Epocast 50-AI + chopped glass fiber;
- soon after, a layer of glass fiber fabric 7781 shall be laminated (E1910363) + resin Epocast 50-AI shall be laminated, with orientation of $\pm 45^\circ$ and overlap of 25 mm, on both sides of the part, on the covered hole;
- the laminate shall be cured for 150 min at 65°C according to PF 15-001; and
- the repair shall be inspected visually according to NE 20-032 requirements, Level III.

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8.6 SUPERFICIAL POROSITY

Superficial porosity repair is allowed in parts according to indicated in NE 20-032.

TABLE 9 : SUPERFICIAL POROSITY

PARTS	DISCREPANCY LIMITATIONS
PARTS OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-500/505 (PHENOM 100/300) AND EMB-314 (ALX)	SUPERFICIAL POROSITY ACCORDING TO NE 20-032

REPAIR:

- apply resin EA9396 or EA9320NA;
- the repair shall be cured for 1 h at 65°C;
- sand the affected surface (sandpapers grit 60 and/to up to 180) and clean with a cloth soaked in solvent specified by NE 40-012; and
- the repair shall be inspected visually according to NE 20-032 requirements, Level III.

8.7 DISBONDING BETWEEN HONEYCOMB AND SOLID LAMINATE

The repair de disbonding between honeycomb and solid laminate is allowed in parts, fulfilling all the limitations indicated in Table 10.

TABLE 10 : DISBONDING BETWEEN HONEYCOMB AND SOLID LAMINATE

PARTS	DISCREPANCY LIMITATIONS
1. CABIN OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 2. NON-STRUCTURAL PARTS OF INTERIORS. 3. SEGREGATION WALL OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 4. FURNISHINGS (DIVIDERS, WARDROBES, TOILETS), CABIN PAX LINERS (SIDE WALL, SIDE LEDGE) AND COCKPIT OF THE PROGRAMS EMB-500/505 (PHENOM 100/300). OBS: NOT APPLICABLE FOR CEILING PANELS, FLOOR PANELS, COCKPIT MAIN PANEL, CONTROL PEDESTAL AND GLARESHIELD OF THE PROGRAMS EMB-500/505 (PHENOM 100/300).	1. MAXIMUM DIMENSION OF 50 x 50 mm. 2. MANUFACTURED PARTS WITH MATERIAL E8353578 (HT93 PHENOLIC), LIMIT OF 5% OF THE PART TOTAL AREA.

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REPAIR:

- a) execute holes of diameter 2,5 mm according to necessity assuring minimum distance between holes (pitch) de 12 mm. The holes may be executed so much in the internal part so external part according to necessity;
- b) inject resin Epocast 50-AI with the effected part returned down so that the gravity force aids the resin flow;
- c) if necessary, the system line may be corrected laminating plies of glass fiber fabric 7781 (E1910363) or glass fiber fabric 116 (E1910379) with resin Epocast 50-AI;
- d) the resin shall be cured for 150 min at 65°C according to PF 15-001;
- e) the finishing in the repair shall be performed with sanding (sandpapers grit 60 and/or up to 180); and
- f) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

8.8 DAMAGE DUE TO TOOL ENTRANCE IN THE PARTS EDGE

The repair of damage due to tool in the parts edge is permitted in parts, fulfilling all the limitations indicated in Table 11.

TABLE 11 : DAMAGE DUE TO TOOL ENTRANCE IN THE PARTS EDGE

PARTS	DISCREPANCY LIMITATIONS
1. CABIN OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 2. NON-STRUCTURAL PARTS OF INTERIORS. 3. SEGREGATION WALL OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 4. FURNISHINGS (DIVIDERS, WARDROBES, TOILETS), CABIN PAX LINERS (SIDE WALL, SIDE LEDGE) AND COCKPIT OF THE PROGRAMS EMB-500/505 (PHENOM 100/300). OBS: NOT APPLICABLE FOR CEILING PANELS, FLOOR PANELS, COCKPIT MAIN PANEL, CONTROL PEDESTAL AND GLARESHIELD OF THE PROGRAMS EMB-500/505 (PHENOM 100/300).	1. MAXIMUM DIMENSION OF 5 x 5 mm. 2. DISTANCE BETWEEN THE DAMAGE AND THE CLOSEST HOLE SHALL BE EQUAL OR SUPERIOR AT 3d . d = diameter of the closest hole. 3. MANUFACTURED PARTS WITH MATERIAL E8353578 (HT93 PHENOLIC), LIMIT OF 5% OF THE PART TOTAL AREA.

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REPAIR:

- the damage shall be filled with resin Epocast 50-A1 + chopped glass fiber;
- soon after, a layer of fabric (according to Table 1) + resin Epocast 50-A1 shall be laminated with orientation of $\pm 45^\circ$ and overlap of 25 mm, on both sides of the part, on the covered hole;
- the laminate shall be cured for 150 min at 65°C according to PF 15-001; and
- the repair shall be inspected visually according to NE 20-032 requirements, Level III.

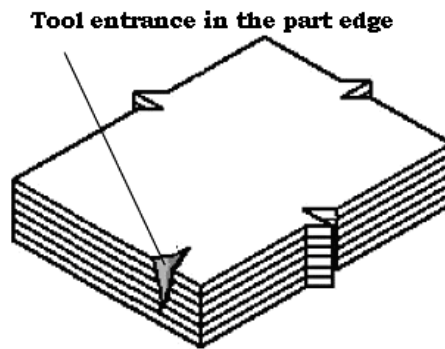


FIGURE 7 : DAMAGE DUE TOOL ENTRANCE IN THE PARTS EDGE

8.9 DISBONDING/DELAMINATION/BREAKING/DISPLACEMENT OR LACK OF THE FIRST OR LAST PLY OF GLASS FIBER IN CARBON FIBER PARTS

The repair of disbonding/delamination/breaking/displacement or lack of the first or last ply of glass fiber in carbon fiber parts is permitted, fulfilling all the limitations indicated in Table 12.

TABLE 12 : DISBONDING/DELAMINATION/BREAKING/DISPLACEMENT OR LACK OF THE FIRST OR LAST PLY OF GLASS FIBER IN CARBON FIBER PARTS

PARTS	DISCREPANCY LIMITATIONS
PARTS OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-500/505 (PHENOM 100/300) AND EMB-314 (ALX).	WHEN THE GLASS FIBER PLY HAS EXCLUSIVELY THE FUNCTION GALVANIC PROTECTION.

REPAIR:

- recompose the layer according to design lay up using glass fiber fabric 116 (E1910379) or glass fiber fabric 7781 (E1910363), using resin Epocast 50-A1;
- the laminate shall be cured for 150 min at 65°C according to PF 15-001, compacting in way to avoid steps; and
- the repair shall be inspected visually according to NE 20-032 requirements, Level III.

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8.10 HONEYCOMB CRUSHING

The repair of honeycomb crushing is allowed in parts, fulfilling all the limitations indicated in Table 13.

TABLE 13 : HONEYCOMB CRUSHING

PARTS	DISCREPANCY LIMITATIONS
1. CABIN OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 2. NON-STRUCTURAL PARTS OF INTERIORS. 3. SEGREGATION WALL OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 4. FURNISHINGS (DIVIDERS, WARDROBES, TOILETS), CABIN PAX LINERS (SIDE WALL, SIDE LEDGE) AND COCKPIT OF THE PROGRAMS EMB-500/505 (PHENOM 100/300). OBS: NOT APPLICABLE FOR CEILING PANELS, FLOOR PANELS, COCKPIT MAIN PANEL, CONTROL PEDESTAL AND GLARESHIELD OF THE PROGRAMS EMB-500/505 (PHENOM 100/300).	1. UP TO 50% OF HONEYCOMB HEIGHT. 2. MANUFACTURED PARTS WITH MATERIAL E8353578 (HT93 PHENOLIC), LIMIT OF 5% OF THE PART TOTAL AREA.

REPAIR:

- a) the system line of the part shall be recomposed with resin L-306-12 or EC-3550 B/A FST (applicable to all programs) or with resin Epocast 1652 (not applicable to the programs EMB-500/505);
- b) the repair shall be cured for 2 h at 65°C when resin Epocast 1652 is applied or for 90 min at 65°C when resin L-306-12 or EC-3550 B/A FST is applied;
- c) apply one ply of glass fiber fabric 7781 (E1910363) and resin Epocast 50-A1 on the reworked area with overlap of 20 mm;
- d) the repair shall be cured for 150 min at 65°C according to PF 15-001; and
- e) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

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8.11 CRACKS IN PARTS WITH HONEYCOMB

Crack repair in parts with honeycomb is allowed in parts, fulfilling all the limitations indicated in Table 14.

TABLE 14 : CRACKS IN PARTS WITH HONEYCOMB

PARTS	DISCREPANCY LIMITATIONS
1. CABIN OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 2. NON-STRUCTURAL PARTS OF INTERIORS. 3. SEGREGATION WALL OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 4. FURNISHINGS (DIVIDERS, WARDROBES, TOILETS), CABIN PAX LINERS (SIDE WALL, SIDE LEDGE) AND COCKPIT OF THE PROGRAMS EMB-500/505 (PHENOM 100/300). OBS: NOT APPLICABLE FOR CEILING PANELS, FLOOR PANELS, COCKPIT MAIN PANEL, CONTROL PEDESTAL AND GLARESHIELD OF THE PROGRAMS EMB-500/505 (PHENOM 100/300).	1. UP TO 300 mm OF LENGTH. 2. FOR PARTS WITH UP TO 2 PLIES. 3. MANUFACTURED PARTS WITH MATERIAL E8353578 (HT93 PHENOLIC), LIMIT OF 5% OF THE PART TOTAL AREA.

REPAIR:

- a) the damage shall be totally eliminated through the sanding (sandpapers grit 60 and/or up to 180) of the affected area, maintaining a stagger of 15 mm (reference);
- b) if the honeycomb is affected, fill with resin Epocast 1652 (E8203537) and cure for 2 h at 65°C;
- c) the original lay-up shall be recomposed with fabric (according to Table 1) and resin Epocast 50-A1 with overlap of 15 mm;
- d) a ply of the same material, with orientation of $\pm 45^\circ$ and with overlap of 20 mm it shall be added on the repair;
- e) the repair shall be cured for 150 min at 65°C according to PF 15-001; and
- f) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

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8.12 DISPLACED INSERTS

The repair of displaced inserts is allowed in parts, fulfilling all the limitations indicated in Table 15.

TABLE 15 : DISPLACED INSERTS

PARTS	DISCREPANCY LIMITATIONS
1. SHELL PANELS OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE). 2. NON-STRUCTURAL PARTS OF INTERIORS. 3. FURNISHINGS (DIVIDERS, WARDROBES, TOILETS), CABIN PAX LINERS (SIDE WALL, SIDE LEDGE) AND COCKPIT OF THE PROGRAMS EMB-500/505 (PHENOM 100/300). OBS: NOT APPLICABLE FOR CEILING PANELS, FLOOR PANELS, COCKPIT MAIN PANEL, CONTROL PEDESTAL AND GLARESHIELD OF THE PROGRAMS EMB-500/505 (PHENOM 100/300).	1. MANUFACTURED PARTS WITH MATERIAL E8353578 (HT93 PHENOLIC), LIMIT OF 5% OF THE PART TOTAL AREA. 2. MINIMUM DISTANCE OF DESIGN EDGE. 3. DISPLACEMENT UP TO 0,5*D OR MINIMUM PITCH DE 3*D. D= diameter of greater hole

REPAIR:

- a) remove the displaced insert, fill the hole with resin L-306-12 or EC-3550 B/A FST (applicable to all programs) or with resin Epocast 1652 (not applicable to the programs EMB-500/505);
- b) the repair shall be cured for 90 min at 65°C when resin L-306-12 or EC-3550 B/A FST is applied or for 2 h at 65°C when resin Epocast 1652 is applied;
- c) apply a ply of glass fiber 7781 (E1910363) and resin Epocast 50-AI with overlapping of 15 mm;
- d) the repair shall be cured for 150 min at 65°C, according to PF 15-001; and
- e) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

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8.13 DISPLACED INSERTS/LOCKED (PROGRAM A1M)

The repair of displaced inserts/locked is allowed in parts, fulfilling all the limitations indicated in Table 15A.

TABLE 15A : DISPLACED INSERTS/LOCKED (PROGRAM A1M)

PARTS	DISCREPANCY LIMITATIONS
APPLICABLE TO THE PANELS PN'S 217A3210-401/402, 217A2069-401/402 AND 217A2067-401/402 OF THE PROGRAM A1M.	1. LIMIT OF 5% OF THE PART TOTAL AREA. 2. MINIMUM DISTANCE OF DESIGN EDGE. 3. DISPLACEMENT UP TO 0,5*D OR MINIMUM PITCH DE 3*D . D = diameter of greater hole.

REPAIR:

- sand with stagger of 10 to 13 mm in the region of insert removal (internal side);
- remove the damaged insert by side unfinished;
- position the new insert and apply the adhesive EA9320NA around the insert until the part surface (protect the thread not to get adhesive);
- cure for 2 h at 65°C;
- sand the resin equaling the surface;
- recompose the lay-up of plies according to drawing using glass fiber fabric 7781 and resin EA9396 with overlap of 10 to 13 mm;
- add 2 plies to the lay-up. The orientation of internal ply is $\pm 45^\circ$ and external ply is $\pm 90^\circ$;
- apply high compacting and cure for 1 h at 65°C;
- inspect the region of rework according to NE 20-032 requirements, Level II; and
- other conditions according to project.

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9. STANDARD PROCEDURES FOR COSMETIC REPAIRS (NON-STRUCTURAL) IN EXTERNAL FAIRING

NOTES

- 1) In electrical bonding areas, ensure repair conditions according to NE 80-060.
- 2) Reconstitute the glass fiber fabric ply 116 when applicable.

9.1 BREAKING OF PLY AND DELAMINATION

The repairs of tool entrance and delamination are allowed in parts, fulfilling all the limitations indicated in Table 16.

TABLE 16 : BREAKING OF PLY AND DELAMINATION

PARTS	DISCREPANCY LIMITATIONS
1. WING/FUSELAGE FAIRING OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX), EMB-500/505 (PHENOM 100/300) AND EMB-390 (KC-390). 2. FAIRING OF VERTICAL EMPENAGE/ HORIZONTAL STABILIZER INCLUDING "DORSAL FIN AND LEADING EDGE SKIN" OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 3. DOORS AND ACCESS COVERS OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX) AND EMB-500/505 (PHENOM 100/300). 4. VERTICAL STABILIZER LEADING EDGE SKIN, DORSAL, VENTRAL FIN, TAIL BOOM AND TAIL CONE OF EMB-500/505 (PHENOM 100/300). 5. PARTS IN GENERAL PRODUCED WITH 177°C (350°F) CURING PHENOLIC RESIN (MEP 15-016). NOTE REPAIR USING ORIGINAL MATERIAL AND CURE AS PER TABLE 2. 6. SPONSON OF THE PROGRAM EMB-390 (KC-390).	1. MAXIMUM DIMENSION OF 50 x 50 mm. FOR EMB-390 (KC-390) MAXIMUM DIMENSION OF 100 x 100 mm. 2. ONLY ONE EFFECTED STRUCTURAL PLY. FOR EMB-390 (KC-390) UNTIL 20% OF AFFECTED STRUCTURAL PLIES. 3. TO ENSURE MINIMUM DISTANCE OF 25,4 mm (1 in) BETWEEN DAMAGE AND ATTACHMENT HOLES OF PART OR DISCONTINUITIES ("CUT OUT").

REPAIR:

- a) the delaminated layer and burrs (in the case of tool entrance) shall be removed with sanding (sandpapers grit 60 and/to up to 180);
- b) the removed layer shall be reconstituted according to Table 2, with design orientation and overlap of 15 mm. Use adhesive film in the interface in accordance with Table 2;
- c) a reinforcement ply of the same fabric shall be added on the area, with orientation of $\pm 45^\circ$ and overlap of 25 mm;
- d) the repair shall be cured for 120°C, according to NE 40-182, and it shall be sanded (sandpapers grit 60 and/or up to 180) to give the finish; and
- e) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

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9.2 EXCESSIVE CLEARANCE BETWEEN PARTS

The repair of excessive clearance is allowed in parts, fulfilling the limitations indicated in Table 17.

TABLE 17 : ALLOWED REPAIR IN CLEARANCES

PARTS	DISCREPANCY LIMITATIONS
1. WING/FUSELAGE FAIRING OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX), EMB-500/505 (PHENOM 100/300) AND EMB-390 (KC-390). 2. DOORS AND ACCESS COVERS OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX) AND EMB-500/505 (PHENOM 100/300). 3. SPONSON OF THE PROGRAM EMB-390 (KC-390).	1. MAXIMUM WIDTH OF 6,0 mm. 2. MAXIMUM LENTH OF 200 mm.

REPAIR:

- the system line shall be recomposed with resin Epocast 1652;
- the repair shall be cured for 2 h at 65°C;
- the repair shall be inspected visually according to NE 20-032 requirements, Level III;
- the repaired part shall be adjusted according to assembly need, so that it is obtained the project geometry;
- one prepreg ply (according to Table 2), orientation of $\pm 45^\circ$ it shall be laminated on the repair. Use adhesive film in the interface in accordance with Table 2. See Figure 9;
- the laminate shall be cured at 120°C, according to NE 40-182; and
- the repair shall be inspected visually according to NE 20-032 requirements, Level III.

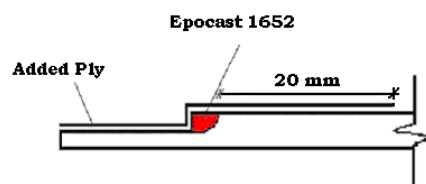


FIGURE 8 : REPAIR OF REGION WITH MATERIAL LACK

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9.3 DELAMINATION IN THE EDGE

The delamination repair in the part edge is allowed in parts, fulfilling all the limitations indicated in Table 18.

TABLE 18 : ALLOWED REPAIR FOR DELAMINATION IN THE EDGE

PARTS	DISCREPANCY LIMITATIONS
1. WING/FUSELAGE FAIRING OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX), EMB-500/505 (PHENOM 100/300) AND EMB-390 (KC-390). 2. FAIRING OF VERTICAL EMPENAGE/HORIZONTAL STABILIZER INCLUDING "DORSAL FIN AND LEADING EDGE SKIN" OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 3. DOORS AND ACCESS COVERS OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX) AND EMB-500/505 (PHENOM 100/300). 4. VERTICAL STABILIZER LEADING EDGE SKIN, DORSAL AND VENTRAL FIN, TAIL BOOM, TAIL CONE OF EMB-500/505 (PHENOM 100/300). 5. PARTS IN GENERAL PRODUCED WITH 177°C (350°F) CURING PHENOLIC RESIN (MEP 15-016). NOTE FOR THIS CLASS OF PARTS, THE USE OF RESIN EA9320 IS NOT PERMITTED. 6. SPONSON OF THE PROGRAM EMB-390 (KC-390).	1. MAXIMUM DIMENSION OF 6 x 50 mm. 2. THE DELAMINATION MAY NOT AFFECT THE ATTACHMENT OF THE PART.

REPAIR:

- inject resin EA9396 or EA9320NA in the delaminated region;
- cure for 1 h at 65°C; compact the region during cure; and
- the repair shall be inspected visually according to NE 20-032 requirements, Level III.

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9.4 DISPLACED OR OUT-OF ROUND HOLE

The repair of displaced or out of round hole is allowed in parts, fulfilling all the limitations indicated in Table 19.

TABLE 19 : DISPLACED OR OUT-OF ROUND HOLE

PARTS	DISCREPANCY LIMITATIONS
1. WING/FUSELAGE FAIRING OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX), EMB-500/505 (PHENOM 100/300) AND EMB-390 (KC-390). 2. FAIRING OF VERTICAL EMPENAGE/HORIZONTAL STABILIZER INCLUDING "DORSAL FIN AND LEADING EDGE SKIN" OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 3. DOORS AND ACCESS COVERS OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX) AND EMB-500/505 (PHENOM 100/300). 4. VERTICAL STABILIZER LEADING EDGE SKIN, DORSAL AND VENTRAL FIN, TAIL BOOM, TAIL CONE OF EMB-500/505 (PHENOM 100/300). 5. PARTS IN GENERAL PRODUCED WITH 177°C (350°F) CURING PHENOLIC RESIN (MEP 15-016). NOTE FOR THIS CLASS OF PARTS, THE USE OF RESIN EA9320 IS NOT PERMITTED. 6. SPONSON OF THE PROGRAM EMB-390 (KC-390).	1. MAXIMUM DISPLACING OF OVAL OR DISPLACED HOLE: 0,5D mm. D = diameter of the greater hole 2. MAXIMUM OF 1 OVAL OR DISPLACED HOLE FOR EACH SEQUENCE OF 10 HOLES.

REPAIR:

- a) the affected hole shall be filled with resin EA9396 or EA9320NA + chopped glass fiber;
- b) in the two sides of the part a ply of glass fiber fabric 7781 (E1910363 shall be laminated with resin EA9396 or EA9320NA;
- c) the laminated part shall be cured for 1 h at 65°C; and
- d) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

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9.5 IMPROPER HOLE OR DRILL MARKS

The repair of improper holes or drill marks is allowed in parts, fulfilling all the limitations indicated in Table 20.

TABLE 20 : IMPROPER HOLE OR DRILL MARKS

PARTS	DISCREPANCY LIMITATIONS
1. WING/FUSELAGE FAIRING OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX), EMB-500/505 (PHENOM 100/300) AND EMB-390 (KC-390). 2. FAIRING OF VERTICAL EMPENAGE/HORIZONTAL STABILIZER INCLUDING "DORSAL FIN AND LEADING EDGE SKIN" OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 3. DOORS AND ACCESS COVERS OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX) AND EMB-500/505 (PHENOM 100/300). 4. VERTICAL STABILIZER LEADING EDGE SKIN, DORSAL AND VENTRAL FIN, TAIL BOOM, TAIL CONE EMB-500/505 (PHENOM 100/300). 5. PARTS IN GENERAL PRODUCED WITH 177°C (350°F) CURING PHENOLIC RESIN (MEP 15-016). NOTE FOR THIS CLASS OF PARTS, THE USE OF RESIN EA9320 IS NOT PERMITTED. 6. SPONSON OF THE PROGRAM EMB-390 (KC-390).	1. MAXIMUM DIAMETER OF IMPROPER HOLE OR DRILL MARK OF: - 5,1 mm (1/5 in) (FOR EMB-500/505) and - 6,4 mm (1/4 in) (FOR OTHER PROGRAMS). 2. MINIMUM DISTANCE OF EDGE OF 2,5d AND PITCH OF 4D mm. d = diameter of improper hole/drill mark D= diameter of greater hole

REPAIR:

- the affected hole shall be filled with resin EA9396 or EA9320NA + chopped glass fiber;
- in the two sides of the part a ply of glass fiber fabric 7781 (E1910363 shall be laminated with resin EA9396 or EA9320NA;
- the laminated part shall be cured for 1 h at 65°C; and
- the repair shall be inspected visually according to NE 20-032 requirements, Level III.

9.6 SUPERFICIAL POROSITY AND SCRATCH

9.6.1 Superficial Porosity

The repair of superficial porosity is allowed in parts as indicated in NE 20-032.

TABLE 21 : SUPERFICIAL POROSITY

SCRATCH PARTS	DISCREPANCY LIMITATIONS
PARTS OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-500/505 (PHENOM 100/300), EMB-550/545 (LEGACY 500/450), EMB-314 (ALX) AND EMB-390 (KC-390).	SUPERFICIAL POROSITY ACCORDING TO NE 20-032.

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REPAIR:

- a) apply resin EA9396 or EA9320NA;
- b) the repair shall be cured for 1 h at 65°C;
- c) execute sanding (sandpapers grit 60 and/to up to 180) and clean with a cloth soaked in solvent specified by NE 40-012; and
- d) the repair shall visually be inspected according to NE 20-032 requirements, Level III.

9.6.2 Superficial Scratch

The repair of superficial scratch is allowed in parts, fulfilling the limitations indicated in Table 22.

TABLE 22 : SUPERFICIAL SCRATCH

PARTS	DISCREPANCY LIMITATIONS
1. WING/FUSELAGE FAIRING OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX), EMB-500/505 (PHENOM 100/300) AND EMB-390 (KC-390). 2. FAIRING OF VERTICAL EMPENAGE/HORIZONTAL ESTABILIZER, INCLUDING "DORSAL FIN AND LEADING EDGE SKIN" OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX) AND EMB-390 (KC-390). 3. DOORS AND ACESS COVER OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX), EMB-500/505 (PHENOM 100/300) AND EMB-390 (KC-390). 4. SPONSON OF THE PROGRAM EMB-390 (KC-390).	SINCE THAT THE FIBER IS NOT EXPOSED OR DAMAGED.

REPAIR:

- a) the part surface shall be sanded with sandpapers of oxide of aluminum grit 180 and clean with a clean cloth soaked in solvent specified by NE 40-012;
- b) in the areas with superficial scratch shall be applied resin EA9396 or EA9320NA;
- c) the resin shall be cured for 1 h at 65°C;
- d) the part surface shall be sanded with sandpapers of oxide of aluminum grit 180 and clean with a clean cloth soaked in solvent specified by NE 40-012; and
- e) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

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9.7 DISBONDING BETWEEN HONEYCOMB AND SOLID LAMINATE

The disbonding repair between honeycomb and solid laminate is allowed in parts, fulfilling the limitations indicated in Table 23.

TABLE 23 : DISBONDING BETWEEN HONEYCOMB AND SOLID LAMINATE

PARTS	DISCREPANCY LIMITATIONS
1. WING/FUSELAGE FAIRING OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX) AND EMB-390 (KC-390). 2. FAIRING OF VERTICAL EMPENAGE/HORIZONTAL ESTABILIZER, INCLUDING "DORSAL FIN AND LEADING EDGE SKIN" OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 3. DOORS AND ACCESS COVER OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 4. SPONSON OF THE PROGRAM EMB-390 (KC-390).	MAXIMUM DIMENSION OF 50 x 50 mm. FOR EMB-390 (KC-390) MAXIMUM DIMENSION OF 100 x 100 mm.

REPAIR:

- execute holes of diameter 2,5 mm according to need guaranteeing minimum distance between holes (pitch) of 12 mm. The holes may be executed so much in the internal part as external part according to need;
- to inject resin EA9396 or EA9320NA with the affected part returned down so that by gravity force aids the resin flow;
- if necessary, the system line may be corrected laminating plies of glass fiber fabric 7781 (E1910363) or glass fiber fabric 116 (E1910379) with resin EA9396 or EA9320NA (use the same resin used in item "b");
- the resin shall be cured for 1 h at 65°C;
- the finish shall be given in the repair with sanding (sandpapers grit 60 and/or up to 180); and
- the repair shall be inspected visually according to NE 20-032 requirements, Level III.

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9.8 DAMAGE DUE TOOL TO ENTRANCE IN THE PARTS EDGE

The repair of damage due to tool entrance in the part edge is allowed in parts, fulfilling all the limitations indicated in Table 24.

TABLE 24 : DAMAGE DUE TOOL TO ENTRANCE IN THE PARTS EDGE

PARTS	DISCREPANCY LIMITATIONS
1. WING/FUSELAGE FAIRING OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX), EMB-500/505 (PHENOM 100/300) AND EMB-390 (KC-390). 2. FAIRING OF VERTICAL EMPENAGE/HORIZONTAL STABILIZER INCLUDING "DORSAL FIN AND LEADING EDGE SKIN" OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 3. DOORS AND ACCESS COVERS OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX) AND EMB-500/505 (PHENOM 100/300). 4. VERTICAL STABILIZER LEADING EDGE SKIN, DORSAL AND VENTRAL FIN, TAIL BOOM, TAIL CONE OF EMB-500/505 (PHENOM 100/300). 5. SPONSON OF THE PROGRAM EMB-390 (KC-390).	1. MAXIMUM DIMENSION OF 5 x 5 mm. FOR EMB-390 (KC-390) MAXIMUM DIMENSION OF 10 x 10 mm. 2. DISTANCE BETWEEN DAMAGE AND THE CLOSEST HOLE SHALL BE THE SAME OR SUPERIOR AT 3d. d = diameter of the closest hole.

REPAIR:

- the damage shall be filled with resin EA9396 or EA9320NA + chopped glass fiber;
- soon after, a fabric layer shall be laminated (according to Table 1) + resin EA9396 or EA9320NA, with orientation of $\pm 45^\circ$ and overlap of 25 mm, on both sides of the part, on the covered hole;
- the laminate shall be cured for 1 h at 65°C; and
- the repair shall be inspected visually according to NE 20-032 requirements, Level III.

Tool entrance in the part edge

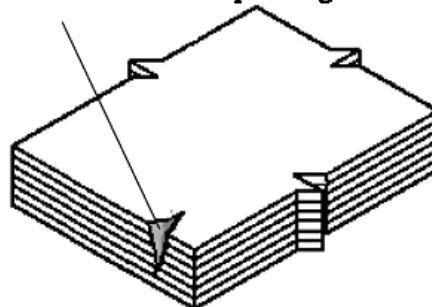


FIGURE 9 : DAMAGE DUE TO TOOL ENTRANCE IN THE PARTS EDGE

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9.9 DISBONDING/DELAMINATION/BREAKING/DISPLACEMENT OR LACK OF THE FIRST OR LAST PLY OF GLASS FIBER IN CARBON FIBER PARTS

The disbonding/delamination/breaking/displacement repair or lack of the first or last ply of glass fiber in carbon fiber parts is allowed, fulfilling the limitations indicated in Table 25.

TABLE 25 : DISBONDING/DELAMINATION/BREAKING/DISPLACEMENT OR LACK OF THE FIRST OR LAST PLY OF GLASS FIBER IN CARBON FIBER PARTS

PARTS	DISCREPANCY LIMITATIONS
PARTS OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-500/505 (PHENOM 100/300), EMB-550/545 (LEGACY 500/450), EMB-314 (ALX) AND EMB-390 (KC-390).	WHEN THE GLASS FIBER PLY HAS EXCLUSIVELY THE FUNCTION GALVANIC PROTECTION.

REPAIR 1: Ply glass fiber fabric disbonded/delaminated/breaked/displaced in relation to the project

- a) recompose the ply in accordance with design using glass fiber fabric 116 (E1910379) or glass fiber fabric 7781 (E1910363), using resin EA9396 or EA9320NA;
- b) the laminate shall be cured for 1 h at 65°C, compacting in way to avoid steps; and
- c) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

REPAIR 2: Lack of glass fiber fabric ply according to design

- a) the ply to be inserted shall be constituted according to Table 2, with design orientation and overlap of 15 mm. Use adhesive film in the interface in accordance with Table 2;
- b) the repair shall be cured at 120°C, according to NE 40-182, and shall be sanded (sandpapers grit 60 and/or up to 180) to give finish; and
- c) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

9.10 LACK, WRINKLE OR BREAKING OF THE TEDLAR PLY

The lack, wrinkle or breaking of Tedlar repair is only allowed in parts, fulfilling the limitations indicated in Table 26.

TABLE 26 : LACK, WRINKLE OR BREAKING OF TEDLAR PLY

PARTS	DISCREPANCY LIMITATIONS
PARTS OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-500/505 (PHENOM 100/300), EMB-550/545 (LEGACY 500/450), EMB-314 (ALX) AND EMB-390 (KC-390).	ONLY THE AFFECTED TEDLAR PLY.

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REPAIR

- a) laminate the Tedlar ply with resin EA9396 or EA9320NA;
- b) apply vacuum bag and cure for 1 h at 65°C; and
- c) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

9.11 LACK, DISBONDING OR BREAKING OF THE TEFLON PLY

The lack, disbonding or breaking of Teflon repair is only allowed in parts fulfilling the limitations indicated in Table 27.

TABLE 27 : LACK, DISBONDING OR BREAKING OF TEFLON PLY

PARTS	DISCREPANCY LIMITATIONS
PARTS OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-500/505 (PHENOM 100/300), EMB-314 (ALX) AND EMB-390 (KC-390).	ONLY THE AFFECTED TEFLON PLY.

REPAIR:

- a) laminate the Teflon ply with adhesive film E2086261;
- b) the laminate shall be cured at 120°C, according to NE 40-182; and
- c) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

9.12 EXCESSIVE ADJUSTMENT IN THE PARTS EDGE

The repair of excessive adjustment is allowed in parts, fulfilling all the limitations indicated in Table 28.

TABLE 28 : ALLOWED REPAIR FOR EXCESSIVE ADJUST

PARTS	DISCREPANCY LIMITATIONS
1. WING/FUSELAGE FAIRING OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX), EMB-500/505 (PHENOM 100/300) AND EMB-390 (KC-390). 2. DOORS AND ACCESS COVERS OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX) AND EMB-500/505 (PHENOM 100/300). 3. VERTICAL STABILIZER LEADING EDGE SKIN, DORSAL AND VENTRAL FIN, TAIL BOOM, TAIL CONE OF EMB-500/505 (PHENOM 100/300). 4. SPONSON OF THE PROGRAM EMB-390 (KC-390).	1. MAXIMUM WIDTH OF 4,0 mm. 2. MAXIMUM LENGTH OF 100 mm. 3. MINIMUM EDGE DISTANCE OF 2,5D , WITHOUT CONSIDERING THE REPAIR. D = diameter of the closest hole to the damage.

REPAIR:

- make just chamfers in the two sides of the laminate in the discrepant areas, according to Figure 10, creating anchorage for the increment of resin EA9396 or EA9320NA + chopped glass fiber, in order to recompose the part system line;
- the repair shall be cured for 1 h at 65°C;
- the piece shall be inspected and adjusted according to assembly need, since its maximum dimensions are according to the design;
- laminate on the two sides one fabric ply and resin (according to Table 1) with orientation of $\pm 45^\circ$, and with overlap of 25 mm;
- the laminate shall be cured for 1 h at 65°C, compacting in way to avoid steps; and
- the repair shall be inspected visually according to NE 20-032 requirements, Level III.

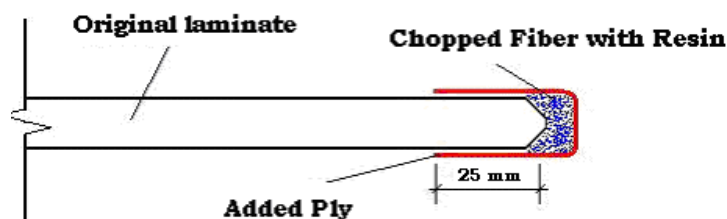


FIGURE 10 : REPAIR OF REGION WITH EXCESSIVE ADJUSTMENT

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9.13 DISPLACEMENT OF HONEYCOMB CAUSING FRICTION IN THE NEXT ASSEMBLIES

The repairs due to the honeycomb displacement causing friction in the next assemblies is allowed in parts, fulfilling all the limitations in Table 29.

TABLE 29 : FRICTION CAUSED BY THE HONEYCOMB DISPLACEMENT

PARTS	DISCREPANCY LIMITATIONS
1. WING/FUSELAGE FAIRING OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX) AND EMB-390 (KC-390). 2. DOORS AND ACCESS COVERS OF PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX). 3. SPONSON OF THE PROGRAM EMB-390 (KC-390).	1. MAXIMUM DISCREPANCY EXTENSION (SOMATORY), OF 500 mm BY PART. 2. ONLY FOR PARTS WHERE THE FLANGE OF ATTACHMENT IS SHORT THAN SPECIFIED IN DESIGN.

REPAIR:

- a) execute sanding (sandpapers grit 60 and/to up to 180) and remove the plies on the honeycomb to be adjusted. Sand in the flange the same number of removed plies on the honeycomb in way to avoid steps in the repair;
- b) adjust the honeycomb according to design;
- c) the removed plies shall be recomposed according to Table 2, with design orientation. Utilize the adhesive film in the rework interface according to Table 2;
- d) one reinforcement ply of the same fabric shall be added on the region, with orientation of $\pm 45^\circ$ and overlapping of 25 mm;
- e) recompose the Tedlar ply according to design;
- f) the repair shall be cured at 120°C , according to NE 40-182, and shall be sanded (sandpapers grit 60 and/to up to 180) to give finish; and
- g) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

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9.14 BREAKING OF PLIES ON THE HONEYCOMB AND/OR HONEYCOMB BREAKING

The repairs due the breaking of plies on the honeycomb and/or honeycomb breaking are allowed in parts fulfilling all the limitations indicated in Table 30.

TABLE 30 : BREAKING OF PLIES ON THE HONEYCOMB AND/OR HONEYCOMB BREAKING

PARTS	DISCREPANCY LIMITATIONS
1. WING/FUSELAGE FAIRING OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX) AND EMB-390 (KC-390). 2. DOORS AND ACESS COVERS OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 3. SPONSON OF THE PROGRAM EMB-390 (KC-390).	MAXIMUM DIMENSION OF 50 x 50 mm. FOR EMB-390 (KC-390) MAXIMUM DIMENSION OF 100 x 100 mm.

REPAIR:

- a) the damage shall be totally eliminated through the sanding (sandpapers grit 60 and/to up to 180) of the affected area, maintaining a stagger of 15 mm (for EMB-390, 20 mm) (reference);
- b) in the case the honeycomb is affected, fill the with resin Epocast 1652 (E8203537) and cure for 2 h at 65°C;
- c) the removed plies shall be recomposed according to Table 2; with design orientation. Utilize adhesive film in the interface according to Table 2;
- d) one reinforcement ply of the same fabric shall be added on the region, with orientation of $\pm 45^\circ$ and overlapping of 25 mm;
- e) recompose the Tedlar ply according to design;
- f) the repair shall be cured at 120°C, according to NE 40-182, and shall be sanded (sandpapers grit 60 and/to up to 180) to give finish; and
- g) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

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9.15 DISPLACEMENT OR IMPROPER HOLES OF ATTACHMENT OF NUT PLATE

The repairs due to the displacement or improper holes of attachment of nut plate are allowed in parts fulfilling all the limitation indicated in Table 31.

**TABLE 31 : DISPLACEMENT OR IMPROPER HOLES
OF ATTACHMENT OF NUT PLATE**

PARTS	DISCREPANCY LIMITATIONS
1. WING/FUSELAGE FAIRING OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX), EMB-500/505 (PHENOM 100/300) AND EMB-390 (KC-390). 2. FAIRING OF VERTICAL EMPENAGE/HORIZONTAL STABILIZER INCLUDING "DORSAL FIN AND LEADING EDGE SKIN" OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE) AND EMB-314 (ALX). 3. DOORS AND ACCESS COVERS OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX) AND EMB-500/505 (PHENOM 100/300). 4. VERTICAL STABILIZER LEADING EDGE SKIN, DORSAL AND VENTRAL FIN, TAIL BOOM, TAIL CONE OF EMB-500/505 (PHENOM 100/300). 5. SPONSON OF THE PROGRAM EMB-390 (KC-390).	1. THE BOLT HOLE MAY NOT BE DISPLACED. 2. HOLE DIAMETER TO BE REPAIRED OF 2,5 mm (3/32in). 3. MINIMUM PITCH BETWEEN THE HOLES ATTACHMENT OF THE FLANGE OF THE NUT PLATE OF 5,00 mm. 4. MINIMUM EDGE DISTANCE OF 2,0*d FOR THE EXECUTED DRILLING.

REPAIR:

- the affected holes shall be filled with resin EA9396 or EA9320NA + chopped glass fiber;
- on the part side shall be laminated one ply of 7781 glass fiber fabric (E1910363) with resin EA9396 or EA9320NA (use the same resin utilized on sub-item "a");
- the laminate shall be cured for 1 h at 65°C;
- the repair shall be inspected visually according to NE 20-032 requirements, Level III;
- install the nut plate of design, rotating the nut plate in way to avoid the spiking of the nut in repaired; and
- ensure that new position of the nut plate flange does not generate impacts in the next assembling.

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9.16 EXTERNAL BRIDGE

The external bridge repair is allowed in parts with acceptance limits indicated in NE 20-032.

REPAIR:

- a) execute sanding (sandpapers grit 60 and/to up to 180) and clean with a cloth soaked in solvent specified by NE 40-012;
- b) the system line shall be recomposed with resin Epocast 1652;
- c) the repair shall be cured for 2 h at 65°C;
- d) execute sanding (sandpapers grit 60 and/to up to 180) and clean with a cloth soaked in solvent specified by NE 40-012; and
- e) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

9.17 SURFACE IMPRESSIONS IN RESIN

The surface impressions in resin repair are allowed in parts with acceptance limits indicated in NE 20-032.

REPAIR:

- a) remove the PTFE film; and
- b) reapply the PTFE film with adhesive film according to Table 2.

9.18 CREASE

The crease repair is allowed in parts with acceptance limits indicated in NE 20-032.

REPAIR:

- a) execute sanding (sandpapers grit 60 and/to up to 180) and clean with a cloth soaked in solvent specified by NE 40-012;
- b) the system line shall be recomposed with resin Epocast 1652;
- c) the repair shall be cured for 2 h at 65°C;
- d) execute sanding (sandpapers grit 60 and/to up to 180) and clean with a cloth soaked in solvent specified by NE 40-012; and
- e) the repair shall be inspected visually according to NE 20-032 requirements, Level III.

9.19 HOLE WITH OVER COUNTERSINK

The repair of over countersink holes is allowed in parts, fulfilling all the limitations indicated in Table 32.

TABLE 32 : OVER COUNTERSINK HOLE

PARTS	DISCREPANCY LIMITATIONS
1. WING/FUSELAGE FAIRING OF THE PROGRAMS EMB-145 (INCLUDING LEGACY), ERJ 170/190 (INCLUDING LINEAGE), EMB-314 (ALX), EMB-500/505 (PHENOM 100/300) AND EMB-390 (KC-390). 2. SPONSON OF THE PROGRAM EMB-390 (KC-390).	1. MAXIMUM ACCEPTABLE DEEP COUNTERSINK IS 2/3 OF THE PART THICKNESS. 2. MAXIMUM OF 1 OVER COUNTERSINK HOLE FOR EACH SEQUENCE OF 10 HOLES. FOR EMB-390 (KC-390) MAXIMUM OF 3 OVER COUNTERSINK HOLES FOR EACH SEQUENCE OF 10 HOLES. 3. REPAIR SHALL NOT BE PERFORMED IF AFFECTS ELECTRICAL BONDING SHEET.

REPAIR:

- lay-up using prepreg materials and adhesive film in interface to fill countersunk according to Table 2 (see Figure 11);
- restore electrical bonding as per NE 80-060 if it is necessary;
- cure the repair at 120°C according to NE 40-182 (cure as per NE 40-183 or NE 40-189 are also allowed);
- execute sanding (sandpapers grit 60 and/to up to 180) to excess remove;
- the repair shall be inspected visually according to NE 20-032 requirements, Class III;
- drill the hole and countersunk as per drawing's requirement; and
- certify remaining conditions as per drawing's requirement.

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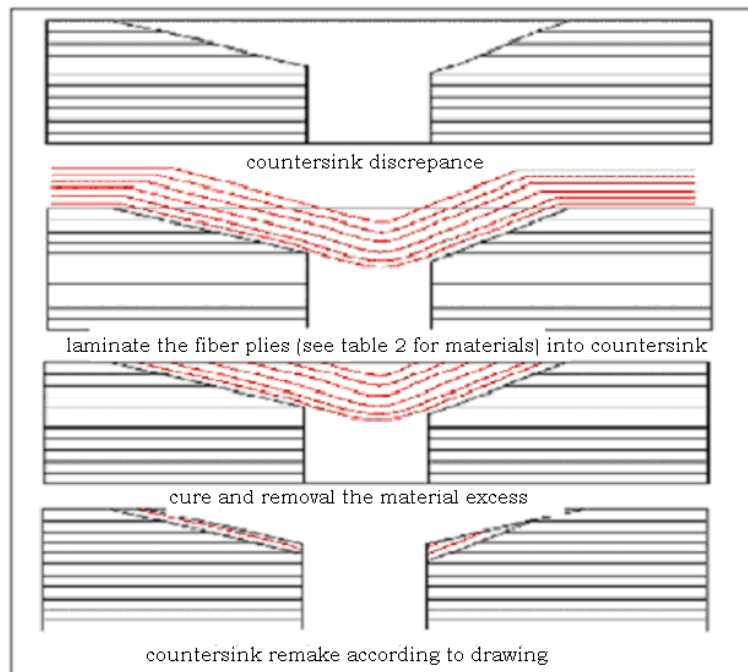


FIGURE 11 : FIBER PLYS LAMINATION

9.20 DISCONTINUITY OF SYSTEM LINE (STEP)

System line deviation, step, caused by tooling partition.

TABLE 33 : DISCONTINUITY OF SYSTEM LINE (STEP)

PARTS	DISCREPANCY LIMITATIONS
1. AIR INLET - EMB314 (ALX), EMB500/505 (PHENOM 100/300), EMB145 (INCLUDING LEGACY), ERJ 170/190. 2. BOOM FAIRING – EMB 500/505 (PHENOM 100/300), EMB 145 (INCLUDING LEGACY). 3. WING TIP FAIRING – EMB 314 (ALX), EMB 500/505 (PHENOM 100/300), EMB 145 (INCLUDING LEGACY). 4. ELEVATOR TIP FAIRING – EMB 314 (ALX), EMB 500/505 (PHENOM 100/300), EMB 145 (INCLUDING LEGACY). 5. RUDDER TIP FAIRING – EMB 314 (ALX). 6. HORIZONTAL STABILIZER TIP FAIRING – EMB 500/505 (PHENOM 100/300), EMB 145 (INCLUDING LEGACY). 7. TAIL CONE – EMB 500/505. 8. VENTRAL TAIL CONE – EMB 505 (PHENOM 300). 9. COCKPIT CONTROL PEDESTAL ERJ 170/190. 10. AIR INTAKE FAIRING – EMB 314 (ALX).	1. REPAIR SHALL NOT BE PERFORMED IF AFFECTS ELECTRICAL BONDING SHEET. 2. DEVIATION > 0,15 mm AND ≤ 0,70 mm.

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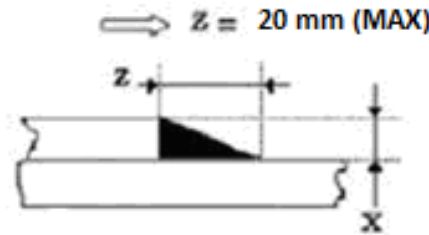
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REPAIR:

- a) sand the surface being careful to not damage the layers;
- b) clean the surface with a cloth soaked in solvent specified by NE 40-012;
- c) apply the **Epocast 7W-A/B** or **Epocast 1652** on the area with the system line deviation (step);
- d) do not apply resin at the higher surface, only at the lower surface;
- e) do not apply excessive resin and do not extend the application area to more than 20 mm (see Figure below):



- f) cure for 2 h at 65°C;
- g) execute sanding to smooth the deviation; and
- h) clean with a cloth soaked in solvent specified by NE 40-012.

10. PATTERN PROCEDURES FOR COSMETIC REPAIRS (NON-STRUCTURAL) IN THE AUXILIARY FUEL TANKS OF LINEAGE 1000

10.1 WATER DETECTION IN THE METALLIC PANELS

If it is detected the water presence in the interior of metallic panels (aluminum-aluminum sandwich), during or by reason of the ultrasonic inspection, procedure as follows:

- a) remove the temporary sealing from all the drains in the extremities;
- b) incline the panel in way that the water is removed by the existing drains in the extremities; if necessary, move the position of the panel;
- c) after the water removal, put the panel in the oven with the drain points still opened to removal of humidity;

NOTE

A vacuum bag in region with water presence shall be used as aspirator bag to improve the water removal and humidity.

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d) dry the panel during 4 h at 60°C;

e) reposition the temporary sealing of all drains in the extremities; and

f) remake the ultrasonic inspection to verify the complete water removal in the panel's interior.

ATTENTION

This procedure may be repeated up to two times. From this point, the Engineering shall be consulted.

11. QUALITY ASSURANCE PROVISIONS

11.1 RECORDS

11.1.1 Special attention shall be given to the records of cure cycles generated in paper (autoclave registration, oven or machine "Hot Bonder") that shall be treated appropriately to guarantee that the information is maintained by the file time of defined according to requirements of the Embraer Quality.

11.1.2 All necessary documentation and registrations to guarantee the accomplishment to the requirements and traceability of the processes shall be ensured through:

- Embraer: Modification Order (MO) or Production Order (PO);
- Subcontractor vendor/Manufacturer: Production Order (PO - of repair); or Service Order (SO); Conformity Discrepancy Registration (CD);
- Visual inspection of the damage and, if necessary, ultrasonic inspection;
- Register the lots of materials used;
- Data of the cure cycle (temperature, vacuum and curing time); and
- Report of conclusion of the repair/rework with photos, when requested by the area of the Engineering or Embraer Quality.

12. BIBLIOGRAPHICAL REFERENCES

SRM for Repairs of Composed Materials.
NCRs of Cosmetic Repairs year 2004.

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